

How Much Does Fusion Design Matter? Partition, Seed, and Cluster-Aware Evaluation of Hybrid LSTM–GAT Traffic Anomaly Detectors

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Abstract

Architectural gains in traffic anomaly detection may depend on data partitions, training seeds, and the statistical treatment of overlapping windows. We evaluate hybrid Long Short-Term Memory–Graph Attention Network (LSTM–GAT) detectors using physics-informed synthetic trajectories over an OpenStreetMap-derived road network. Nine encoder configurations, four fusion operators, and two graph constructions are compared under a controlled multi-seed protocol. Trip-level cluster bootstrap, sign-flip permutation tests, and Holm correction account for window dependence and multiple comparisons. No comparison on the fixed 14-trip partition remains significant after trip-level inference and Holm correction. Under five-fold trip-grouped cross-validation over 90 trips, fusion operators span 0.0205 in mean run-level F1, below within-configuration standard deviations of 0.04–0.05; none is statistically distinguishable from another. Sample-wise gating ranks last on the fixed partition and first by mean run-level F1 under cross-validation. A seed-specific gate collapse recurs in an independent rerun but is not observed with the alternative graph construction. Compared with LSTM-only, static weighted-sum LSTM–GAT improves abnormal-dwell detection by 0.0335 F1 (95% confidence interval [0.0139, 0.0610]; Holm $p = 0.002$), but provides no benefit on speed-limit violations. These findings establish no general fusion-operator advantage and highlight the need for multiple partitions and appropriate statistical-unit inference.

Keywords: traffic anomaly detection; spatio-temporal modeling; long short-term memory; graph attention networks; multimodal fusion; ablation study; grouped cross-validation; cluster bootstrap; permutation tests

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1. Introduction

Intelligent Transportation Systems (ITS) have become a fundamental component of modern urban mobility management by enabling large-scale monitoring and analysis of traffic conditions through heterogeneous data sources. These systems generate massive volumes of spatio-temporal traffic data, providing new opportunities for understanding driving behavior and enhancing road safety. Traffic anomalies—such as sudden stops

or abnormal maneuvers—can disrupt traffic flow and create significant safety risks. Accurate and timely detection is therefore critical for decision-support mechanisms in ITS environments.

Recent studies have leveraged deep learning, particularly **Long Short-Term Memory (LSTM)** networks, to model sequential vehicle trajectories. However, road segments form complex graph-structured networks, making purely temporal reasoning insufficient for context-sensitive anomaly detection. To capture structural dependencies, **Graph Attention Networks (GAT)** and related graph-based approaches have been increasingly adopted alongside temporal encoders. As reflected in recent literature, current research increasingly moves toward both richer spatio-temporal anomaly modeling and more effective fusion strategies for integrating temporal behavior with spatial context [1–4]. Existing approaches frequently suffer from “**contextual blindness**,” where a model flags natural behavior as anomalous due to limited structural awareness, while many multimodal methods still rely on static or weakly adaptive fusion that cannot fully respond to changing traffic conditions [2,4].

A recurring methodological pattern accompanies this progress. Architectural contributions in this area are commonly evaluated through a *single* training run on a *single* train/test partition, and improvements of one to three F1 points are reported as evidence that a proposed fusion or attention mechanism is responsible for the gain. Two properties of the anomaly-detection setting make such evidence fragile. First, anomaly prevalence is low, so the number of positive test instances is small and paired significance tests have limited power. Second, graph message passing on GPU is not bit-reproducible—scatter-based aggregation performs floating-point summation in a nondeterministic order—so repeated runs of an identical configuration do not return identical numbers. Under these conditions, a single-run difference cannot distinguish an architectural effect from seed and partition variance.

This paper therefore asks a narrower and more answerable question than “is the proposed architecture better?”: *which components of a hybrid LSTM–GAT anomaly detector actually carry the observed performance, and how much of the reported advantage survives a controlled protocol?* We keep the task formulation, dataset, grouped partition, optimizer, epoch budget, checkpoint criterion, and threshold-selection rule fixed, vary one component at a time, and report every result over multiple seeds with interval estimates and paired tests. The system under study is the hybrid design we previously proposed, denoted **STANet**, whose fusion stage is a sample-wise scalar modality gate; the evaluation, however, is deliberately symmetric and does not privilege that design.

The main contributions are:

- **Controlled encoder benchmark.** Nine encoder configurations—covering the core operators of STGCN, DCRNN, and Graph WaveNet adapted to the present task—are compared under an identical protocol over five seeds. LSTM–GAT has the highest observed mean F1, while the LSTM, GRU, and Transformer configurations lie within 0.0015 F1 of one another. However, none of the eight encoder comparisons remains significant after trip-level testing and Holm correction (Section 4.8, Table 16). The fixed-partition ranking is therefore descriptive and establishes neither a spatial-operator nor a temporal-encoder advantage.
- **Direct fusion ablation.** Four fusion operators—sample-wise gating, fixed concatenation, static weighted summation, and multi-head cross-attention—are compared with all other components frozen. Sample-wise gating ranks last on the fixed partition and exhibits a degenerate collapse mode in one of five seeds. Separately, the single global mixing coefficient learned by the static weighted-sum variant converges to approximately equal weighting (≈ 0.50) across seeds; this observation does not describe the

sample-wise gate. Repeating the ablation on a second graph construction localizes the collapse: it occurs when the gate locks onto a spatial branch that carries almost no standalone signal, and disappears when that branch is made more informative.

- **Evidence on evaluation protocol.** We show that an ablation conclusion drawn from one fixed grouped partition is not reproduced under five-fold trip-grouped cross-validation covering all sequences. We further show that the windows used as observations overlap by 80%, so the window-level tests customary in this literature overstate the effective sample size; architectural conclusions therefore rely on trip-level cluster bootstrap and permutation tests with Holm correction. Window-level tests are retained in selected tables for comparison, not as the basis for architectural claims.
- **Reference points and transparency.** A zero-parameter threshold-rule classifier derived from the generation thresholds, gradient-boosting baselines on window aggregates, probability calibration, and single-window latency on GPU and CPU are reported alongside the learned models, together with a complete data card for the synthetic dataset. Every result file carries a protocol hash, and configurations appearing in several tables are trained once and shared, so the tables are consistent by construction rather than by assertion.

2. Related Work

The literature most relevant to this study covers both anomaly-detection models and the fusion strategies used to integrate temporal and spatial information.

2.1. Temporal, Spatial, and Spatio-Temporal Anomaly Models

Early traffic anomaly detection studies were predominantly driven by temporal modeling, where recurrent architectures such as LSTM-based networks were used to capture abrupt variations in velocity, acceleration, and short-term driving behavior. These models remain effective when anomalies are primarily reflected in kinematic changes, but they often overlook whether a maneuver is contextually plausible for the surrounding road environment. In parallel, graph-based and spatially aware methods improved structural understanding by modeling road connectivity, segment interactions, and topological constraints. Such approaches are better suited to representing where risky conditions may emerge, yet they can miss the sequential behavioral evolution that distinguishes a routine slowdown from a truly hazardous event.

Recent studies increasingly bridge these two views through spatio-temporal learning. Huang et al. [1] combine Dynamic GAT and Informer components to capture long-range temporal dependencies together with adaptive relational structure. Duan et al. [2] extend this direction through a multimodal deep-fusion framework with heterogeneous graph neural networks. Nguyen et al. [3] further show that graph-based spatio-temporal integration can effectively represent coupled speed–volume patterns in urban traffic environments. Taken together, these studies confirm that anomaly detection benefits from combining behavioral evolution with structural context; how much weight each modality should receive, however, is usually settled by architectural configuration rather than measured directly.

2.2. Fusion Strategies and Context Integration

A second research direction focuses on how information from temporal and spatial branches should be fused. Liang et al. [4] adopt a lightweight spatio-temporal architecture that emphasizes compact feature fusion, whereas Cui et al. [5] use a spatial–temporal attention design for lane-aware trajectory prediction. These studies show that attention and feature-level coupling improve representation quality, yet the relative contribution of

temporal and spatial cues is still largely determined by architectural configuration rather than by an explicit, empirically validated comparison against simpler fusion operators.

The mechanism examined in this paper is a **sample-wise scalar modality gate** applied after the LSTM and GAT outputs have been projected into a shared 64-dimensional space. It produces one scalar per sample and broadcasts that value across all latent features; it is not a feature-level attention vector, a time-step attention mechanism, or a graph-neighborhood coefficient. Table 1 places this mechanism among related designs. A central empirical question of this study is whether that modality-level distinction translates into a measurable advantage; Section 4.3 reports that, under a controlled protocol, it does not.

Beyond fusion design, transfer between simulated and real operating conditions remains a question for the present simulation-trained detectors. Evaluating adaptation to field-collected trajectories and external road networks is a future direction discussed in Section 5.3.

Table 1. Summary of related spatio-temporal anomaly detection methods

Study	Year	Core Arch.	Temporal	Spatial	Fusion
Huang et al. [1]	2024	Dynamic GAT-Informer	Informer	Dynamic GAT	Integrated gated feature fusion with graph-neighborhood attention
Duan et al. [2]	2025	Multimodal deep fusion + HGNN	Traffic flow / speed dynamics	Heterogeneous GNN	Integrated multimodal feature fusion
Nguyen et al. [3]	2026	Spatiotemporal prediction model	Time-series analysis	Graph-based dependency modeling	Integrated spatio-temporal feature fusion
Liang et al. [4]	2025	Lightweight ST network	TCN / CNN	Spatio-temporal features	Compact feature-level fusion
Cui et al. [5]	2026	Lane-flow fusion attention	Temporal self-attention	Lane-level spatial attention	Feature-level attention fusion
This study	2026	Hybrid LSTM-GAT, evaluated component-wise	Stacked LSTM (vs. GRU, gated TCN, dilated conv, Transformer)	GAT (vs. Chebyshev, diffusion, adaptive adjacency, none)	Four operators compared directly: gated, concat, weighted sum, cross-attention

3. Materials and Methods

3.1. Problem Formulation

Urban traffic systems can be modeled as complex spatio-temporal processes in which vehicle movements evolve while simultaneously occurring within the structural constraints of a road network [6,7]. In this study, the traffic anomaly detection problem is formulated as a context-aware learning task that jointly models vehicle trajectory dynamics and road network topology [8]. The mathematical notations used throughout this formulation are summarized in Table 2.

Table 2. Summary of mathematical notation

Symbol	Description
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Directed graph representing the road network
$\mathcal{V}, \mathcal{E}$	Set of nodes (road segments) and edges (connectivity)
$N, \mathbf{A}$	Number of nodes and adjacency matrix encoding connectivity
$\mathbf{s}_i, \mathcal{S}$	Structural feature vector of node i and the set of all structural features
$\mathcal{T} = \{\mathbf{x}_1, \dots, \mathbf{x}_\tau\}$	Vehicle trajectory sequence of length τ
$\mathbf{x}_t, \mathbf{X} \in \mathbb{R}^{\tau \times 4}$	Per-step kinematic vector and stacked temporal feature matrix
v_t, a_t, θ_t, id_t	Velocity, acceleration, heading angle, and road-segment ID at time t
x_{sin}, x_{cos}	Trigonometric heading components used for continuous angular representation
$type_i, length_i$ $deg_i, context_i$	Road category and segment length; node degree and contextual attributes
$\mathbf{h}_t, \mathbf{h}_{temp}, \mathbf{h}_{spat}$	LSTM hidden state, temporal embedding, and spatial embedding
$\Theta_{temp}, \Theta_{spat}$	Learnable parameter sets of the LSTM and GAT branches
$\alpha_{ij}, \mathcal{N}_i$	Graph-attention coefficient from node j to node i and neighborhood of node i
$\mathbf{W}, \vec{a}, \mathbf{h}'_i$	GAT transform matrix, attention vector, and updated node representation
d	Shared latent dimension used before multimodal fusion
$\mathbf{h}'_{temp}, \mathbf{h}'_{spat}$	Projected temporal and spatial embeddings in shared latent space
$\mathbf{W}_t, \mathbf{b}_t$ $\mathbf{W}_s, \mathbf{b}_s$	Learnable projection parameters for temporal/spatial branches
$g_{gate}, \mathbf{W}_g, b_g$	Sample-wise scalar modality weight and gating-network parameters
$\mathbf{z}$ $f_{LSTM}(\cdot), f_{GAT}(\cdot)$ $f_{fusion}(\cdot), g(\cdot)$	Joint spatio-temporal fused embedding Temporal encoder, spatial encoder, fusion, and classification mappings
$\sigma(\cdot)$	Activation function (sigmoid where specified)
$\mathbf{y}, \hat{\mathbf{y}}$ $y_i, \hat{y}_i$	Ground-truth and predicted labels (global and sample-level forms)
$\mathcal{L}, M$	Loss function and mini-batch size
w_{pos}	Positive-class weight in the WBCE loss
Φ	Overall mapping function for anomaly detection

3.1.1. Road Network Representation

The road network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a set of N nodes (road segments) and $\mathcal{E}$ denotes the set of edges representing physical connectivity [9,10]. Each node $v_i \in \mathcal{V}$ is associated with a structural feature vector $\mathbf{s}_i$:

$$\mathbf{s}_i = [type_i, length_i, deg_i, context_i] \quad (1)$$

where $type_i$ denotes the road category, $length_i$ is the segment length, deg_i is the node degree, and $context_i$ represents additional intersection characteristics [11,12]. Spatial dependencies are encoded via an adjacency matrix $\mathbf{A}$ [13,14].

3.1.2. Vehicle Trajectory Modeling

Vehicle movement is represented as a trajectory sequence $\mathcal{T} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_\tau\}$, where τ is the sequence length and each observation $\mathbf{x}_t$ describes the kinematic state at time t [15,16]:

$$\mathbf{x}_t = [v_t, a_t, \theta_t, id_t] \quad (2)$$

where v_t is speed, a_t is acceleration, θ_t is the heading angle, and id_t is the identifier of the road segment where the vehicle is located.

3.1.3. Spatio-Temporal Feature Representation

Anomaly detection requires the integration of behavioral dynamics and structural context [17,18]. We define the learning task as finding a joint embedding $\mathbf{z}$:

$$\mathbf{z} = f(\mathbf{h}_{temp}, \mathbf{h}_{spat}) \quad (3)$$

where $\mathbf{h}_{temp} = \text{LSTM}(\mathcal{T})$ and $\mathbf{h}_{spat} = \text{GAT}(\mathcal{G}, \mathbf{A})$ [19,20]. The function $f(\cdot)$ denotes the fusion operator, which is treated in this study as an *experimental variable* rather than a fixed design choice.

3.1.4. Traffic Anomaly Detection Objective

The detection task is formulated as a binary classification:

$$\hat{y} = \sigma(g(\mathbf{z})) \quad (4)$$

where $g(\cdot)$ is the classification head and $\hat{y} \in [0, 1]$. At the task-formulation level:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^M [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

where $y_i \in \{0, 1\}$ is the ground truth label and M is the number of samples [2,21,22]. Equation 5 is the generic task-level formulation only; because the training partition is imbalanced, the implemented objective is the weighted binary cross-entropy in Eq. 20.

3.1.5. Problem Definition

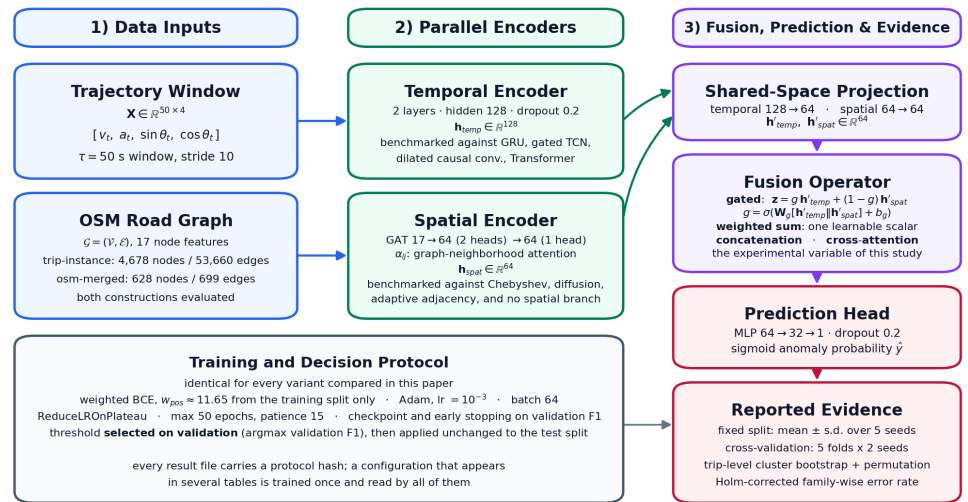
Given a road network $\mathcal{G}$, vehicle trajectory $\mathcal{T}$, and road features $\mathcal{S}$, the objective is to learn a mapping $\Phi : \{\mathcal{T}, \mathcal{G}, \mathcal{S}\} \rightarrow \{0, 1\}$ [8,23].

3.2. Architecture Under Study

This section describes the hybrid architecture whose components are evaluated in Section 4. The model jointly encodes temporal vehicle dynamics and spatial road-network topology through parallel branches and combines them with a configurable fusion operator. We describe the mechanism here and defer all claims about its effectiveness to the experimental sections; the protocol governing those experiments is given in Section 3.4.

3.2.1. Overall Architecture

As illustrated in Fig. 1, the framework consists of (i) a temporal module for sequence modeling, (ii) a spatial module for structural graph learning, and (iii) a fusion module that combines the two projected representations.



Terminology: α_{ij} weights GAT neighbors within the graph; g weights the complete temporal and spatial embeddings against each other. The two are distinct mechanisms.

Figure 1. Overall architecture. The GAT neighborhood coefficients α_{ij} are distinct from the sample-wise scalar modality gate g_{gate} ; the fusion block is the experimental variable of Section 4.3.

Given a vehicle trajectory sequence $\mathcal{T}$ and the road network graph $\mathcal{G}$, the model learns two complementary latent embeddings:

$$\mathbf{h}_{temp} = f_{LSTM}(\mathcal{T}; \Theta_{temp}) \quad (6)$$

$$\mathbf{h}_{spat} = f_{GAT}(\mathcal{G}, \mathbf{A}; \Theta_{spat}) \quad (7)$$

where Θ_{temp} and Θ_{spat} denote the learnable parameter sets of the two branches [20,24]. These embeddings are integrated through the fusion operator:

$$\mathbf{z} = f_{fusion}(\mathbf{h}_{temp}, \mathbf{h}_{spat}) \quad (8)$$

The fused representation $\mathbf{z}$ is then mapped by a classification head to the anomaly probability $\hat{y}$.

3.2.2. Temporal Module

Kinematic Feature Representation

The temporal branch processes four kinematic features at each time step t :

$$\mathbf{x}_t = [v_t, a_t, \sin(\theta_t), \cos(\theta_t)] \quad (9)$$

Representing the heading angle θ_t using its trigonometric components prevents periodic discontinuities at the angular boundaries [16]. Each driving window is represented as $\mathbf{X} \in \mathbb{R}^{\tau \times 4}$ with $\tau = 50$.

LSTM-Based Sequence Encoding

A two-layer stacked LSTM encoder is used [15,16,25]:

$$\mathbf{h}_t = \text{LSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}) \quad (10)$$

Each layer contains 128 hidden units with dropout 0.2 between recurrent layers. The final hidden state of the second layer is exported as the temporal embedding, $\mathbf{h}_{temp} = \mathbf{h}_\tau$. Table 3 summarizes the branch. Section 4.2 replaces this encoder with four alternatives while keeping everything else fixed.

Table 3. Summary of LSTM branch configuration

Component	Setting	Role / Output
Input window	$\tau = 50, 4$ features	Tensor shape $[B, 50, 4]$
Input features	$v_t, a_t, x_{sin}, x_{cos}$	Driving-profile representation
Stacked LSTM	input size = 4, hidden size = 128	Temporal dependency learning
Number of layers	2	Hierarchical sequence encoding
Dropout	$p = 0.2$	Regularization between LSTM layers
Temporal embedding	$\mathbf{h}_{temp} \in \mathbb{R}^{128}$	Output tensor $[B, 128]$
Parameters	200,704	89.2% of the full model

3.2.3. Spatial Module

Graph-Based Contextual Features

The spatial module uses a GAT to encode the road network structure [6,9,12]. Each node v_i is associated with a 17-dimensional structural vector $\mathbf{s}_i$ capturing road-type descriptors, topological metrics (betweenness and closeness centrality), and contextual constraints (traffic-light indicators, tunnel information, rarity scores).

Graph Attention Mechanism

The GAT layers compute adaptive attention coefficients α_{ij} over adjacent road segments [8,19,26]:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\vec{a}^T [\mathbf{W}\mathbf{s}_i \parallel \mathbf{W}\mathbf{s}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\vec{a}^T [\mathbf{W}\mathbf{s}_i \parallel \mathbf{W}\mathbf{s}_k]))} \quad (11)$$

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}\mathbf{s}_j \right) \quad (12)$$

Two consecutive GAT layers are used: the first maps the 17-dimensional node attributes to a 64-dimensional intermediate representation (32 hidden channels, 2 attention heads); the second produces the final 64-dimensional spatial embedding. The embedding of the road segment occupied by the vehicle is extracted as $\mathbf{h}_{spat}$. Table 4 summarizes the branch.

Table 4. Summary of GAT branch configuration

Component	Setting	Role / Output
Graph input	4,678 nodes, 53,660 edges	Trip-specific segment instances
Distinct physical roads	628 unique <code>osm_segment_id</code>	See Section 3.3.2
Node features	17 dimensions	Structural and contextual descriptors
GAT layer 1	GATConv(17, 32, heads= 2)	Intermediate tensor $[N, 64]$
GAT layer 2	GATConv(64, 64, heads= 1)	Spatial embedding tensor $[N, 64]$
Spatial embedding	$\mathbf{h}_{spat} \in \mathbb{R}^{64}$	Output of active road segment
Parameters	5,568	2.5% of the full model

3.2.4. Fusion Module

Since $\mathbf{h}_{temp}$ and $\mathbf{h}_{spat}$ originate from different branches, they are first projected into a common latent dimensionality $d = 64$:

$$\mathbf{h}'_{temp} = \text{ReLU}(\mathbf{W}_t \mathbf{h}_{temp} + \mathbf{b}_t) \quad (13)$$

$$\mathbf{h}'_{spat} = \text{ReLU}(\mathbf{W}_s \mathbf{h}_{spat} + \mathbf{b}_s) \quad (14)$$

Four fusion operators are considered. The **sample-wise scalar modality gate** originally proposed computes Eq. 15 and applies it in Eq. 16:

$$g_{gate} = \sigma(\mathbf{W}_g[\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}] + b_g) \quad (15)$$

$$\mathbf{z} = g_{gate} \cdot \mathbf{h}'_{temp} + (1 - g_{gate}) \cdot \mathbf{h}'_{spat} \quad (16)$$

where $g_{gate} \in [0, 1]$ has tensor shape $[B, 1]$: exactly one scalar per sample, broadcast across all 64 dimensions. The three comparison operators are **fixed concatenation** ($\mathbf{z} = [\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}]$, no reweighting), **static weighted summation** ($\mathbf{z} = s \mathbf{h}'_{temp} + (1 - s) \mathbf{h}'_{spat}$ with a single global learnable scalar s), and **multi-head cross-attention** (four heads, temporal embedding as query, both embeddings as key/value).

We note a structural relation that motivates the ablation. If g_{gate} converges to a constant $\bar{g}$, then Eq. 16 followed by a linear classifier $\mathbf{W}$ yields $\mathbf{Wz} = \bar{g} \mathbf{W} \mathbf{h}'_{temp} + (1 - \bar{g}) \mathbf{W} \mathbf{h}'_{spat}$, which fixed concatenation can reproduce exactly by setting its two weight blocks to $\bar{g} \mathbf{W}$ and $(1 - \bar{g}) \mathbf{W}$. Concatenation therefore spans a strictly larger function class than constant-gate fusion, and any advantage of gating must come from genuinely sample-dependent behavior. Section 4.4 tests whether such behavior is observed.

The fused representation passes through a two-layer MLP ($64 \rightarrow 32 \rightarrow 1$, ReLU, dropout 0.2) to produce the anomaly probability. Table 5 summarizes the stage.

Table 5. Summary of the fusion stage and the four operators compared

Component	Setting	Fusion parameters
Temporal projection	Linear $128 \rightarrow 64 + \text{ReLU}$	—
Spatial projection	Linear $64 \rightarrow 64 + \text{ReLU}$	—
Gated (proposed)	Linear $128 \rightarrow 32 \rightarrow 1 + \text{sigmoid}$	4,161
Fixed concatenation	$[\mathbf{h}'_{temp} \parallel \mathbf{h}'_{spat}]$	0
Static weighted sum	single learnable scalar s	1
Cross-attention	4-head, $d = 64$	16,640
Classifier	MLP $64/128 \rightarrow 32 \rightarrow 1$	—

3.3. Dataset and Preprocessing

3.3.1. Data Sources

The empirical evaluation relies on a **physics-informed synthetic vehicle trajectory dataset** and **OpenStreetMap (OSM)** road network topology.

Vehicle Trajectory Dataset

No pre-existing real-world source dataset was used. To obtain trajectory data with fully known ground-truth anomaly labels—difficult to obtain at scale from real sensor logs because of privacy constraints, sparse anomaly incidence, and the absence of verified incident annotations—this study employs a physics-informed synthetic trajectory-generation framework. Vehicle motion is simulated over the OSM-derived Kocaeli road network using a Markov-chain behavioral model in which each vehicle transitions between four kinematic states (constant speed, acceleration, deceleration, stop) at one-second resolution. State-transition probabilities are conditioned on the local traffic regime (free-flow, transition, congested), defined through a log-normal traffic-intensity distribution and a speed–density relationship adapted from Pipes’ car-following model [27]:

$$\bar{v}_s = (v_{\max} - 0.08v_{\max}) \left(1 - \frac{\tau}{10}\right)^4, \quad (17)$$

where $v_{\max}$ is the segment's speed limit and $\tau \in [1, 9]$ is the sampled traffic-intensity level. Acceleration is constrained to $[-3, 1.5]$ m/s² under normal driving, following AASHTO geometric-design guidance [28]. Sensor-level noise is injected into the speed and bearing channels at every time step.

The generated dataset contains 293,168 records across 90 inter-district trips. Ground-truth labels are recorded at injection time according to the taxonomy in Table 7. A complete data card—generation parameters, subtype rules, imputation shares, and known limitations—is released with the code and data (Section 5.3).

OpenStreetMap Road Network Data

Each road segment is represented by a 17-dimensional feature vector $\mathbf{s}_i$ [12] comprising structural indicators (lane count, road category), infrastructure elements (traffic lights, tunnels), and topological measures (betweenness and closeness centrality).

Several structural attributes required imputation because of incomplete OSM tagging. Lane counts and speed limits, when missing, were assigned the road-type-conditional defaults in Table 6; assigned speed values were additionally smoothed using

$$\hat{v}_i = \begin{cases} \text{Round}_5\left(\frac{v_{i-1} + v_i + v_{i+1}}{3}\right), & C_i, \\ v_i, & \text{otherwise,} \end{cases} \quad (18)$$

$$C_i \equiv (S_i \neq \text{OSM}) \wedge (|v_{i-1} - v_i| > 30),$$

to remove physically implausible discontinuities. Traffic-light presence was inferred using a hybrid rule combining OSM `traffic_signals` tags with a node-degree heuristic.

The values in Table 6 are **author-defined imputation assumptions**, not observed OSM attributes or statutory limits. **53.8% of lane counts and 35.3% of speed limits originate from OSM**; the remainder are imputed defaults (46.2%) or smoothed values (14.8% of speeds). This is material for interpreting the spatial branch, because $v_{\max}$ is one of its node features.

Table 6. Author-defined road-type-conditional imputation defaults

Road Type	Default Lanes	Default Speed (km/h)
motorway	3	130
motorway_link	1	80
trunk	2	110
trunk_link	1	60
primary	2	90
primary_link	1	50
secondary	1	80
secondary_link	1	40
(default)	–	50

The network covers 10 central districts of Kocaeli, Türkiye, with 90 inter-district routes forming the trip set.

3.3.2. Graph Construction and Node Multiplicity

The graph is constructed by treating each *trip-specific traversal* of a road segment as a separate node. This yields 4,678 nodes and 53,660 directed edges, with an average node degree of 11.47. It is important to state the structure this reflects: the 4,678 nodes correspond to only **628 distinct physical roads** identified by `osm_segment_id`, so each physical road

appears on average 7.4 times, once per trip that traverses it. Node features are identical across the copies of a given physical road in 538 of 546 multi-copy groups.

Merging the copies into a single node per physical road yields a substantially different graph: 628 nodes, 699 edges, and an average degree of 1.11. The reported degree of 11.47 therefore describes the trip-instance graph, not the underlying road network, which is closer to a set of overlapping route corridors than to a densely connected urban network.

Either construction imposes a ceiling on what the spatial branch alone can achieve. Because several sequences with different labels map to the same node, road identity alone bounds achievable accuracy at 95.13% on the trip-instance graph and 93.45% on the merged graph. The spatial branch can therefore act as a context provider but cannot serve as a standalone classifier—a prediction confirmed in Section 4.2.

Which construction is primary, and on what grounds

We adopt the trip-instance graph as the primary setting and report the merged graph alongside it. The criterion is *per-node label purity*, which is a property of the data and is computed before any model is trained: distributing sequences over 4,678 nodes rather than 628 raises the label-consistency ceiling from 93.45% to 95.13%, because fewer conflicting labels are forced onto a shared node. We state this explicitly because the alternative justification—selecting the construction that produces the better downstream score—would make the graph choice a function of the outcome it is used to support. The trade-off runs the other way as well and should be weighed by the reader: the merged graph is the faithful representation of the physical road network, and duplicating a road once per traversal inflates the reported average degree and weakens message passing between genuinely adjacent roads. Both graphs are evaluated under the identical protocol in Section 4.7, and the conclusions of this paper are reported for both.

3.3.3. Data Cleaning and Feature Engineering

Records with implausible sensor readings were removed before sequence construction. Speed values outside $[0, 200]$ km/h were excluded ($n = 60$), and road-gradient values outside $[-50\%, 50\%]$ were excluded ($n = 8,185$). In total, 8,245 records (2.81%) were removed, yielding 284,923 records:

$$\text{retain row } i \iff 0 \leq v_i \leq 200 \wedge -50 \leq r_i \leq 50. \quad (19)$$

The raw heading angle is transformed into sine and cosine components to prevent discontinuities at the $0^\circ/360^\circ$ boundary [16].

3.3.4. Anomaly Taxonomy and Label Generation

Anomalies are injected during simulation according to the two-category, nine-subtype taxonomy in Table 7. 6.86% of simulated rows carry a non-zero anomaly label; after windowing, 7.07% of sequences are anomalous.

Table 7. Anomaly taxonomy, generation rules, and observed shares

No.	Subtype	Generation Rule	Rows
1	Speed-limit violation	$v_t > 1.10 v_{\max}$	1,405
2	Abnormal dwell	Stationary for > 30 s	11,610
3	Abnormal acceleration	$a_t \in [-6, -3] \cup [1.5, 2.5] \text{ m/s}^2$	299
4	Accident	Composite perturbation + prolonged stop	5,741
5	Abnormal heading change	$ \Delta\theta_t > 10^\circ$	213
6	GPS speed artifact	Perturbation from kinematic history	66
7	GPS acceleration artifact	Perturbation from kinematic history	76
8	GPS heading artifact	Perturbation from kinematic history	94
9	Signal freeze	Identical values for ≥ 2 s	606

Label–input dependence. Subtypes 1, 2, 3, 5, and 9 are threshold functions of the same kinematic channels supplied to the temporal branch. This is a property of the dataset, not an artifact of the models, and it is quantified directly in Section 4.1 through a zero-parameter rule classifier. One subtype is an informative exception: speed-limit violation depends on $v_{\max}$, which is *not* an input to the temporal branch and is available only as a graph node feature. Section 4.5.1 tests whether the spatial branch exploits this.

3.3.5. Feature Normalization, Sequences, and Partitioning

Velocity is normalized with Min–Max scaling and acceleration with Z-score standardization. **Scalers are fitted on the training partition only** and applied to validation and test data, avoiding the distributional leakage that arises when scaling precedes partitioning.

Trajectories are transformed into fixed-length sequences using a sliding window of $\tau = 50$ steps with a stride of 10, yielding 28,090 sequences of which 7.07% are anomalous. The window label and road segment are taken from the final time step.

As summarized in Section 3.3.5, the dataset is partitioned by unique trip identifiers rather than by individual samples, so that overlapping windows from the same route cannot appear in more than one partition. Two evaluation regimes are used:

- a **fixed grouped partition** (70/15/15 by trip) used for the encoder and fusion comparisons, summarized in Table 8;
- **five-fold trip-grouped cross-validation**, in which every one of the 28,090 sequences receives an out-of-fold test prediction, used in Section 4.5.

Table 8. Fixed grouped partition and anomaly distribution

Partition	Trips	Sequences	Anomaly Ratio
Train	63	19,255	7.90%
Validation	13	4,036	7.58%
Test	14	4,799	3.27%

The lower test prevalence (3.27% versus 7.90% in training) arises from the anomaly composition of the selected whole trips and was not corrected by moving windows across partitions, which would violate the grouping constraint. This imbalance is one motivation for the cross-validated regime, where fold-wise test prevalence ranges from 3.40% to 10.49%.

3.4. Experimental Protocol

3.4.1. Controlled Comparison Design

Every comparison in this paper varies **one component at a time** while holding the following fixed: task formulation, dataset, grouped partition, optimizer (Adam, learning rate 0.001), batch size (64), maximum epochs (50), early-stopping patience (15), learning-rate schedule (*ReduceLROnPlateau*, factor 0.5, patience 3), gradient clipping (max-norm 1.0), loss function, checkpoint-selection criterion, and threshold-selection rule. Encoder comparisons additionally hold the fusion operator fixed, and fusion comparisons hold both encoders fixed.

Loss Function

Because anomalies are rare, a **weighted binary cross-entropy** objective is used:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^M [w_{pos} \cdot y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (20)$$

with $w_{pos} = N_{neg}^{train} / N_{pos}^{train} \approx 11.65$, computed exclusively from the training partition after the grouped split.

Checkpoint and Threshold Selection

A single criterion—best validation F1-score—governs both checkpoint retention and early stopping for *all* configurations. Mixing criteria across variants (for example, early-stopping on loss while checkpointing on F1) makes runs incomparable and is avoided.

For individual-run metrics, the decision threshold is selected on the **validation** partition by maximizing validation F1 over a grid, and then applied unchanged to the test partition. Selecting a threshold on held-out validation data is standard practice and does not constitute test leakage; fixing it a priori at 0.50, as in the previous version of this work, discards performance without any methodological benefit, because a model trained with $w_{pos} = 11.65$ is not calibrated to that operating point.

Repeatability and Reporting

Every learned configuration on the fixed partition is trained with the same five random seeds (42–46), and all results are reported as **mean ± standard deviation**. The cross-validated analysis uses five trip-grouped folds crossed with two seeds, giving ten observations per configuration.

Multi-seed reporting is not optional in this setting. Graph message passing uses scatter-based aggregation, whose floating-point summation order is nondeterministic on GPU; enabling deterministic cuDNN algorithms does not remove this nondeterminism, and we measured two runs of an identical configuration and seed differing by 0.025 F1. Bit-level reproducibility is therefore unattainable, and a single reported number carries an unquantified error bar.

Protocol Stamping and Shared Runs

Because bit-level reproducibility is unattainable, the burden falls on *procedural* consistency: any two numbers presented as comparable must have been produced under exactly the same protocol. An earlier version of this study did not enforce that, and it failed in a way worth reporting. Halfway through the experiment campaign the cuDNN determinism flag was switched off—deterministic algorithms were making the dilated convolution backward pass roughly forty times slower while providing no reproducibility guarantee, for the reason just given—and the experiments run before and after that change were subsequently placed in the same tables. The configuration LSTM + GAT + weighted

sum consequently appeared with two different scores in two tables of the same manuscript, with nothing in the record to explain the discrepancy.

Two mechanisms now prevent this. First, each experiment records a *protocol stamp*: a hash over the seed set, epoch budget, patience, batch size, learning rate, scheduler parameters, dropout, checkpoint criterion, threshold rule, cuDNN flags, window length, stride, split seed, and library versions. Scripts that import another experiment's results assert equality of stamps and abort on mismatch, so results from different protocols cannot silently enter one table. Second, a configuration that appears in more than one table is trained *once* and cached under its protocol stamp; the tables then read the same run. In this paper LSTM + GAT + weighted sum appears in four tables, and all four report the identical cached run rather than four independently trained runs presumed to agree. Changing any protocol field changes the hash, invalidates the cache, and forces retraining.

3.4.2. Unit of Statistical Analysis

Windows are extracted with length $\tau = 50$ and stride 10, so two consecutive windows share 40 of their 50 time steps—an overlap of 80%—and all windows drawn from one trip share a driver, a vehicle, and a route. Windows are therefore *not* independent observations, and paired tests that treat them as such (window-level McNemar, window-level bootstrap) overstate the effective sample size: confidence intervals come out too narrow and p -values too small. This is a property of the sampling design, not of any particular model, and it applies to every comparison in this paper.

Architectural conclusions are based on tests using the **trip** as the statistical unit, with Holm correction within the stated comparison families. This does not mean that every reported p -value is trip-level: window-level results are retained for the fixed-partition baseline, encoder, and fusion comparisons to illustrate the effect of treating dependent windows as independent. Section 4.8 and Table 16 reassess those comparisons over the 14 fixed-test trips. In contrast, the cross-validated fusion comparisons in Section 4.5 and the subtype comparisons in Table 14 already use trip-level inference over all 90 trips. Seed- or run-level means and standard deviations are descriptive summaries, not independent-window significance tests. The trip-level procedures are:

- **Cluster bootstrap.** Trips, not windows, are resampled with replacement; a selected trip contributes all of its windows. Confidence intervals are the 2.5th and 97.5th percentiles of 5,000 resampled F1 differences.
- **Cluster permutation test.** Under the null hypothesis that two models are exchangeable within a trip, an independent coin flip per trip swaps the two models' predictions on all windows of that trip. The two-sided p -value is the proportion of 20,000 such relabelings whose F1 difference is at least as extreme as the observed one. This is the valid counterpart of McNemar's test for clustered data.
- **Multiplicity.** Within each family of comparisons (baselines, fusion operators, encoder configurations) Holm–Bonferroni step-down correction controls the family-wise error rate at $\alpha = 0.05$.

Both quantities are computable in closed form over per-trip contingency counts, since F1 depends only on the totals TP, FP, and FN and these are additive over trips; the implementation was verified against a naive resampling loop.

The consequence is a real loss of statistical power, and we state it plainly rather than absorbing it silently. The dataset contains 90 trips in total; the fixed test partition contains 14 trips, which are the independent units used in the cluster-level analysis. This is the total test-trip count, not a count restricted to trips containing positive windows. Interval estimates on that partition are correspondingly wide. Under cross-validation every one of the 90 trips receives an out-of-fold prediction, which is why the cross-validated results are

treated as the primary evidence in this paper and the fixed-partition results as a secondary, more variable view of the same question. Window-level values are retained in the fixed-partition result tables and accompanying result files, labelled as such, so that readers can see the magnitude of the difference the correct unit makes.

3.4.3. Evaluation Metrics

Performance is assessed using accuracy, precision, recall, F1-score, ROC-AUC, average precision, and—for the first time in this line of work—probability calibration via the Brier score and Expected Calibration Error (ECE).

3.4.4. Implementation

Experiments were conducted with PyTorch 2.7.1 (CUDA 11.8) and PyTorch Geometric 2.7.0 on an NVIDIA RTX 4060 Laptop GPU (8 GB). All preprocessing, model, training, and evaluation code, together with the complete result files, is released (Section 5.3).

4. Results and Discussion

4.1. Reference Points: Rule-Based and Classical Baselines

Because several anomaly subtypes are defined as threshold functions of the model’s own input channels (Table 7), we first establish how much of the task is solvable without learning, and how much is solvable with classical feature-based learning. Table 9 and Fig. 2 report both alongside LSTM–GAT with static weighted-sum fusion. The hybrid reference in Tables 9 and 10 is this configuration, not the sample-wise gated variant. All learned models use the fixed grouped partition, corrected normalization, validation-selected thresholds, and multi-seed reporting.

Table 9. Reference points on the fixed test partition ($n = 4,799, 157$ positives). The LSTM–GAT hybrid uses static weighted-sum fusion.

Model	Precision	Recall	F1-Score	ROC-AUC
Rule: $v < 1$ only (0 params)	—	—	0.8129	—
Rule: $v < 1 \vee a > 3$ (0 params)	—	—	0.8317	—
Rule: 3 thresholds (0 params)	0.8354	0.8726	0.8536	—
XGBoost (window aggregates)	0.9200	0.8166	0.8651 ± 0.0052	0.9966
LightGBM (window aggregates)	0.9355	0.8127	0.8698 ± 0.0075	0.9966
Random Forest (window aggr.)	0.9518	0.8013	0.8699 ± 0.0061	0.9964
LSTM only (no spatial branch)	0.9400	0.7694	0.8455 ± 0.0190	0.9294
LSTM–GAT (weighted sum)	0.9336	0.8178	0.8713 ± 0.0188	0.9638

Three observations follow. First, a **zero-parameter classifier using three generation thresholds reaches** $F1 = 0.8536$, within 2 points of a 220,802-parameter neural model. Second, gradient-boosting models on 38 hand-crafted window aggregates reach 0.865–0.870 and attain **higher ROC-AUC than any neural configuration** (0.9966 versus 0.9638), indicating better threshold-independent ranking. Third, the hybrid model attains the highest mean F1, but by a margin of 0.0014 over Random Forest—roughly one tenth of its own seed-to-seed standard deviation.

Table 10 reports window-level diagnostics rather than the basis for architectural conclusions. The fixed test partition contains 14 trips in total and 157 positive windows (Section 3.4.2); these are counts of different units, not 157 independent positive observations. None of the baseline comparisons evaluated at the trip level remains significant after Holm

correction (Table 16); however, that analysis uses the dwell-only rule, not the three-threshold rule. A trip-level comparison against the latter is not reported in the archived results, so its window-level result cannot establish either superiority or equivalence. The cross-validated analysis of Section 4.5 provides evidence across partitions.

Table 10. Window-level paired comparisons against LSTM–GAT with static weighted-sum fusion, with uncorrected McNemar p -values. The two $\Delta F1$ columns are *different quantities* and are reported separately: the first is the difference of five-seed means, the second is the difference on the median seed, which is the run the interval and the test are computed on. The interval is the quantity that brackets the median-seed difference, not the mean.

Comparison	$\Delta F1$ (5-seed mean)	$\Delta F1$ (median seed)	Bootstrap 95% CI (median seed)	McNemar p
vs. LightGBM	+0.0016	+0.0192	[−0.0063, +0.0473]	0.302
vs. XGBoost	+0.0063	+0.0280	[+0.0034, +0.0558]	0.049
vs. Random Forest	+0.0014	+0.0084	[−0.0132, +0.0303]	0.754
vs. 3-threshold rule	+0.0178	+0.0316	[−0.0086, +0.0740]	0.072
vs. LSTM only	+0.0258	+0.0081	[−0.0147, +0.0316]	0.549

Two corrections to the previous version of this table are worth recording, because both produced numbers that could not be reconciled with each other. First, the mean difference and the interval were placed in one row without stating that they are computed on different quantities—the mean over seeds and a single seed respectively—so a mean falling outside its neighbouring interval looked like an arithmetic error. Second, and more consequentially, the paired bootstrap applied the *hybrid model's* decision threshold to the competitor's probabilities. Since the models are calibrated differently, this evaluated every competitor away from its own operating point and inflated the difference systematically in the hybrid's favour: the Random Forest interval read [+0.038, +0.116] against a mean difference of +0.0137. With each model scored at its own validation-selected threshold, the same interval is [−0.0132, +0.0303], and the comparison is not significant. The values above use the corrected procedure. The intervals and tests remain window-level diagnostics. For comparators included in both analyses, conclusions use the trip-level results in Section 4.8.

The rule comparisons require a separate provenance clarification. In the archived v2.0.0 results, `F1b_fair.json` evaluates the three-threshold rule_best comparator, with a window-level McNemar $p = 0.0724543$ (rounded to 0.072 in Table 10). By contrast, `F7_cluster_stats.json` evaluates rule_dwell_only, with window-level $p = 6.96017 \times 10^{-5}$, trip-level $p = 0.06525$, and Holm-adjusted $p = 0.31973$. These are different rule comparators. The latter two values must not be attributed to the three-threshold rule, whose trip-level comparison is absent from that archive.

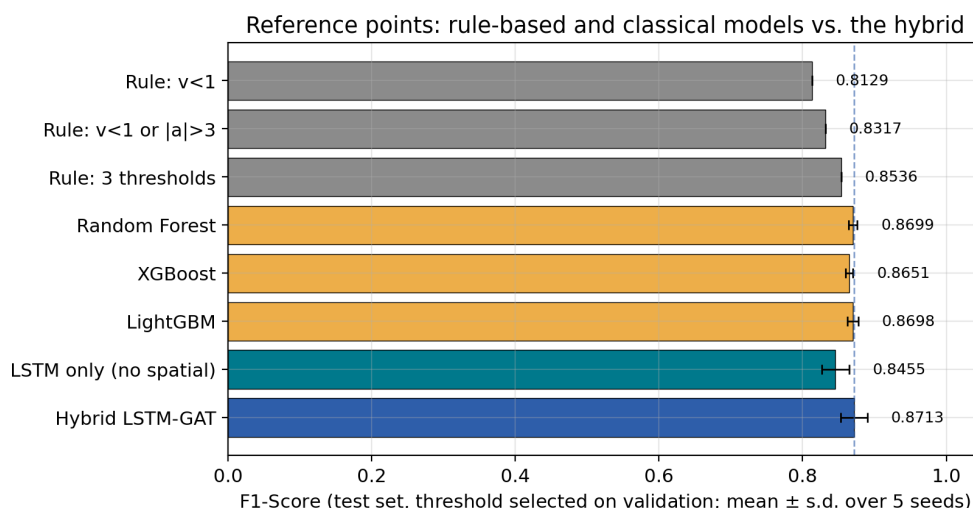


Figure 2. Reference points: zero-parameter rules and classical feature-based models against the static weighted-sum LSTM–GAT hybrid on the fixed test partition.

4.2. Encoder Benchmark

Reviewers of spatio-temporal anomaly detectors reasonably ask for comparison against STGCN [29], DCRNN [30], and Graph WaveNet [31]. These architectures target a structurally different formulation: every graph node carries a continuous time series and the model predicts jointly across all nodes at each step. The present task maps a single vehicle’s 50-step window to a single road segment for binary classification, so these models cannot be deployed unmodified.

We therefore isolate the **core operator** of each architecture and evaluate it under an identical task, budget, partition, fusion operator, and classification head. On the temporal axis we compare the stacked LSTM against a GRU (the DCRNN recurrent backbone), a gated temporal convolution (STGCN), a dilated causal convolution stack with residual and skip connections (Graph WaveNet), and a Transformer encoder. On the spatial axis we compare the GAT against Chebyshev graph convolution (STGCN), bidirectional diffusion convolution (DCRNN), a learnable adaptive adjacency (Graph WaveNet), and no spatial branch at all. This adaptation is a deliberate scoping decision and is stated as such.

Table 11 and Fig. 3 report the outcome. The fusion operator is held at static weighted summation throughout this benchmark. This choice follows from Section 4.3: the gated operator exhibits a collapse mode that would inject variance unrelated to encoder quality. The table retains window-level p -values solely for comparison with the trip-level reassessment in Section 4.8; inferential conclusions are based on the Holm-corrected trip-level values in Table 16.

Table 11. Fixed-partition encoder benchmark over five seeds under an identical protocol. F1 is reported as mean $\pm$ seed standard deviation. The p column contains uncorrected window-level values, not the basis for inferential conclusions. No encoder comparison is significant after trip-level Holm correction (Table 16).

Config	Origin	F1-Score	ROC-AUC	Params	p (window)
LSTM + GAT	this study	0.8713 $\pm$ 0.0188	0.9638	220,802	—
GRU	DCRNN backbone	0.8701 $\pm$ 0.0102	0.9929	170,626	0.119
Transformer	Transformer	0.8698 $\pm$ 0.0081	0.9899	292,098	0.815
Dilated causal conv	Graph WaveNet	0.8535 $\pm$ 0.0114	0.9799	416,258	0.115
No spatial branch	control	0.8455 $\pm$ 0.0190	0.9294	215,234	0.549
Gated TCN	STGCN	0.8314 $\pm$ 0.0058	0.9861	122,242	0.003
Chebyshev conv	STGCN	0.8234 $\pm$ 0.0389	0.9481	223,106	0.007
Diffusion conv	DCRNN	0.8181 $\pm$ 0.0450	0.9633	228,290	0.003
Adaptive adjacency	Graph WaveNet	0.4657 $\pm$ 0.2833	0.7891	321,850	< 0.001

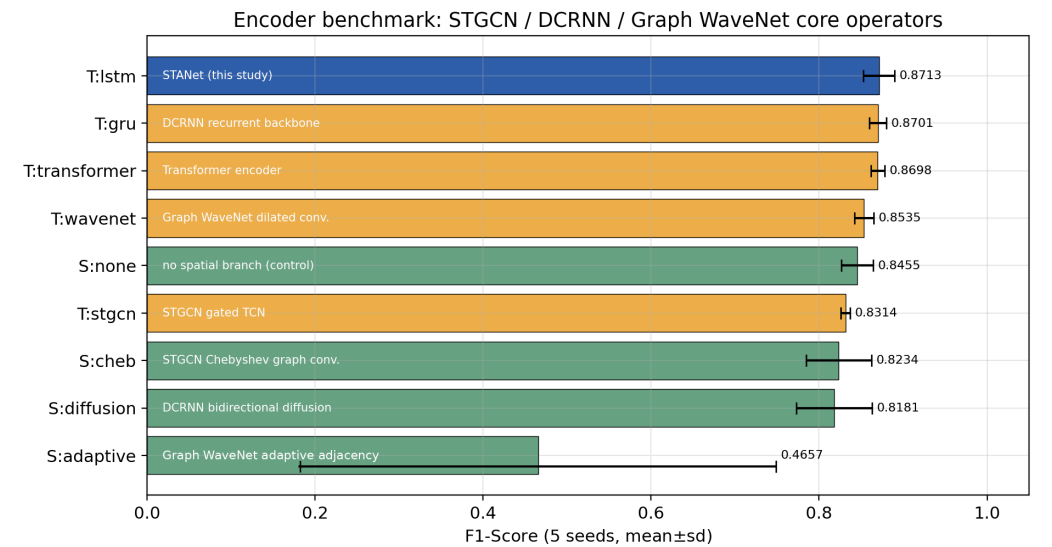


Figure 3. Encoder benchmark. Bars show mean F1 over five seeds; error bars show standard deviation.

The LSTM–GAT pairing attains the highest observed mean F1 among the nine configurations. Window-level tests yield $p = 0.003$ against the gated TCN, $p = 0.007$ against Chebyshev convolution, $p = 0.003$ against diffusion convolution, and $p < 0.001$ against adaptive adjacency. These values treat overlapping windows as independent and do not establish superiority under the adopted inferential protocol. In the trip-level reassessment, the corresponding Holm-corrected values are 0.335, 0.365, 0.365, and 0.056, respectively (Table 16); none meets the 0.05 significance threshold. The remaining four encoder comparisons likewise do not survive correction. The LSTM–GAT, GRU, and Transformer configurations have mean F1 values of 0.8713, 0.8701, and 0.8698, respectively, spanning 0.0015, compared with the LSTM–GAT seed standard deviation of 0.0188. These are descriptive performance differences, not a statistically established ordering.

Accordingly, **neither spatial-operator nor temporal-encoder superiority is established by the fixed-partition encoder comparisons after trip-level Holm correction.** In particular, the observed GAT ranking does not justify a claim that attention-based aggregation is superior to the Chebyshev, diffusion, or adaptive alternatives. Nor does the absence of significance demonstrate equivalence between operators. The GRU configuration has a similar observed F1, 23% fewer parameters, and the highest observed ROC-AUC in this

encoder benchmark; these descriptive trade-offs may motivate further evaluation but do not establish an overall preferable architecture.

Two descriptive observations merit further investigation. The Chebyshev and diffusion configurations have lower observed mean F1 than the no-spatial-branch control, while adaptive adjacency has low mean performance and high seed variability (0.4657 ± 0.2833). These observations do not establish that any benefit of graph message passing is specific to attention-based aggregation, nor do they identify the cause of the adaptive-adjacency instability. Together with the conditional gate failure described in Section 4.3, they motivate additional stability experiments rather than a general claim that minimally parameterized alternatives are superior.

Note on the previous version

The earlier three-seed benchmark reported 0.8836 ± 0.0033 for LSTM-GAT and window-level significance against the DCRNN backbone and both Graph WaveNet components. Extending to five seeds raised the LSTM-GAT standard deviation from 0.0033 to 0.0188, and the GRU and dilated-convolution comparisons no longer met the window-level significance threshold. More importantly, none of the eight current encoder comparisons is significant after trip-level Holm correction. The seed extension illustrates uncertainty in estimates of variability; it does not validate the remaining window-level significance claims.

4.3. Fusion Ablation

We now hold both encoders fixed and vary only the fusion operator, over five seeds. Table 12 and Fig. 4 summarize the outcome.

Table 12. Fusion ablation, five seeds, all other components identical

Fusion	F1-Score	ROC-AUC	Worst seed	Params
Static weighted sum	0.8713 ± 0.0188	0.9638	0.8491	1
Cross-attention	0.8559 ± 0.0228	0.9417	0.8369	16,640
Fixed concatenation	0.8442 ± 0.0091	0.8993	0.8339	0
Gated (proposed)	0.6953 ± 0.3488	0.8752	0.0715	4,161

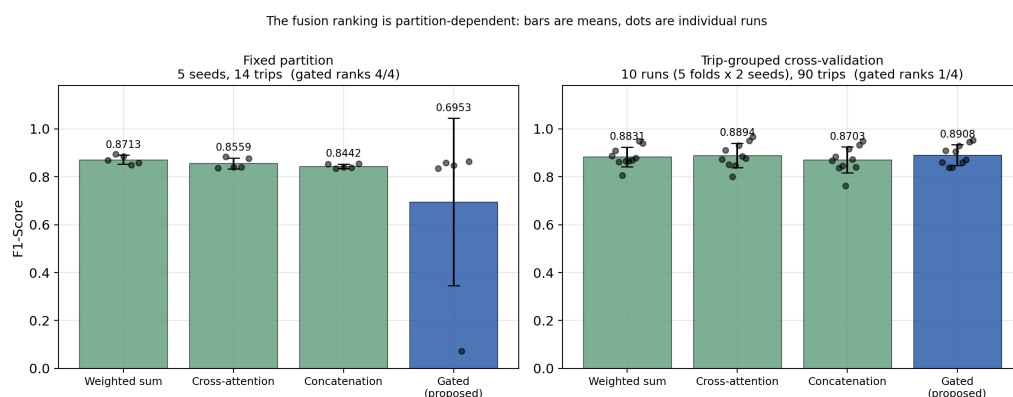


Figure 4. The fusion ranking is partition-dependent. Left: the fixed partition (five seeds, 14 trips), where the gated operator ranks fourth of four and one run collapses. Right: trip-grouped cross-validation (ten runs, 90 trips), where the same operator ranks first of four and no run collapses. Bars are means with standard deviation; dots are individual runs. Under cross-validation no pairwise difference is significant at the trip level after Holm correction (Section 4.5).

The sample-wise gate ranks **last of four**. Two aspects of this result matter more than the ranking itself.

A degenerate collapse mode. In one of five seeds the gated model failed completely ($F1 = 0.0715$), with a mean g_{gate} of 0.026 and a mean of 0.0125 on anomalous samples. The gate had locked onto the spatial modality, which—as Section 4.2 shows—cannot classify on its own, and training could not escape. Equation 16 admits this optimum by construction: as $g_{gate} \rightarrow 0$ the temporal signal is suppressed multiplicatively and its gradient vanishes with it. Concatenation has no such mode, because the classifier weights the two embeddings independently and can simply ignore the uninformative one. No collapse occurred in any of the fifteen runs of the other three operators.

This collapse is reproducible in the strongest sense available to us. The earlier version of this study observed it at seed 45 under a different cuDNN configuration ($F1 = 0.0833$). The experiments reported here were re-run from scratch under the protocol of Section 3.4.1, and the collapse recurred *at the same seed* with a comparable magnitude. It is a property of the operator and this initialization, not an artifact of a particular backend setting.

The learned optimum is a constant. In the static weighted-sum variant, the single learnable scalar converged to $\{0.519, 0.522, 0.518, 0.498, 0.497\}$ across the five seeds—that is, to approximately equal weighting in every run. The data does not call for a sample-dependent mixing weight. A one-parameter static weight outperforms the 4,161-parameter adaptive gate, and does so with an order of magnitude lower seed variance (± 0.0188 versus ± 0.3488).

Paired single-run tests are the wrong instrument for this comparison. On the median seed the gated model is indistinguishable from concatenation ($p = 1.000$) and from cross-attention ($p = 0.727$), and its difference from the static weighted sum is -0.0379 with interval $[-0.0708, -0.0119]$; none of these single-partition numbers reflects the 0.35 standard deviation that is the substantive finding. Across-seed instability is not visible to a test that examines one seed. This is itself an argument for the multi-seed protocol of Section 3.4.1.

4.4. Gate Behavior

Section 3.2.4 established that gating can only outperform concatenation through genuinely sample-dependent behavior. Fig. 5 examines whether such behavior is observed, by measuring the class-conditional shift $\bar{g}_{\text{anomalous}} - \bar{g}_{\text{normal}}$ across seeds.

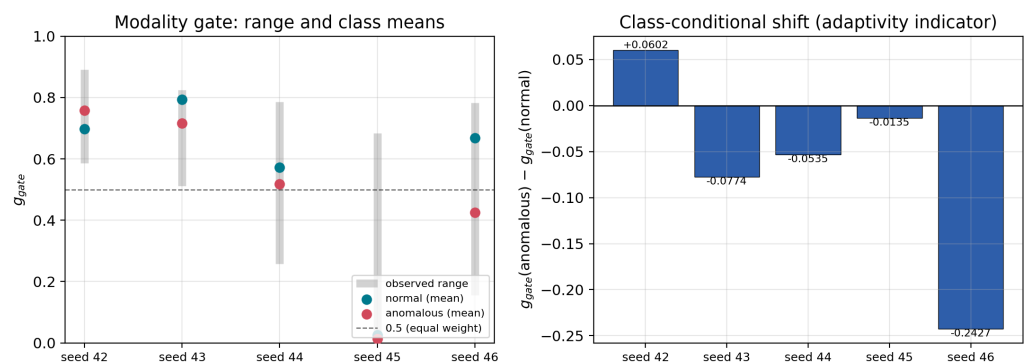


Figure 5. Modality gate behavior across five seeds. Left: observed range and class means. Right: class-conditional shift, which is inconsistent in both magnitude and sign.

The measured shifts are $+0.0602$, -0.0774 , -0.0535 , -0.0135 , and -0.2427 . Both the magnitude and the *sign* vary across seeds. A class-conditional interpretation of the gate—for instance, that anomalous samples shift weight toward the temporal branch—is therefore not a stable property of the trained model but an observation that holds in one run and reverses in the other four.

This is a correction to the interpretation offered in the earlier version of this work, which reported a shift from 0.70–0.75 on normal samples to approximately 0.80 on anomalous samples based on a single checkpoint. Re-evaluating the archived checkpoint gives class means of 0.860 and 0.856 respectively—a difference of 0.004, with the gate never falling below 0.5 on any sample. We report this explicitly because interpretability claims derived from one training run are exactly the kind of evidence this study argues against.

4.5. Cross-Validated Evaluation and Subtype Stratification

The fixed partition of Table 8 contains 157 positive test instances drawn from 14 trips. The trip-level reassessment reported in Section 4.8 finds no significant fixed-partition comparison after Holm correction. We therefore repeat the entire evaluation—all four fusion operators and the spatial-branch control—under five-fold trip-grouped cross-validation crossed with two seeds, giving ten runs per configuration. Every sequence receives an out-of-fold test prediction, so all **90 trips** enter the analysis rather than 14, and fold-wise test prevalence ranges from 3.40% to 10.49%. This is the primary evidence base of the paper.

Within-fold validation split. The archived v2.0.0 implementation in `stanet/data.py` applies five-fold GroupKFold to the unique trip IDs without shuffling, holding out 18 trips per fold. From the remaining 72 trips, `train_test_split` selects validation trips with a validation fraction of 0.1765, shuffling, and split seed $42 + k$ for fold index $k = 0, \dots, 4$; no label stratification is applied. This yields 59 training, 13 validation, and 18 test trips in every fold, as recorded in `F5_cv.json`. All windows from a trip remain together. The split is shared across configurations and training seeds 42 and 43. Normalization is fitted only on the training trips; validation is used for checkpoint and run-specific threshold selection, not taken from the held-out test fold.

Run summaries versus pooled analysis. Run F1, worst-run F1, and collapse counts summarize all ten fold–seed runs. In `experiments/f5_cv.py`, pooling instead concatenates only the five seed-42 test-probability vectors: each of the 28,090 windows occurs once, without seed averaging or duplication. For each configuration, the pooled decision threshold is the arithmetic mean of its ten validation-selected thresholds, including both seeds. Thus, pooled F1 uses one configuration-specific threshold across folds, rather than each run’s own threshold. The recorded pooled thresholds are 0.791 (weighted sum), 0.761 (gated), 0.716 (concatenation), 0.719 (cross-attention), and 0.668 (LSTM-only). Brier score and ECE use the seed-42 pooled probabilities without thresholding. Fusion and subtype tests use these pooled predictions and thresholds, retaining trip IDs for cluster resampling. Their intervals are conditional on this pooling and threshold procedure; they do not measure between-seed uncertainty.

Table 13. Five-fold trip-grouped cross-validation crossed with two seeds (ten runs per configuration, all 28,090 sequences). Collapse counts runs with $F1 < 0.30$. Run summaries use both seeds; pooled F1, Brier, and ECE use seed-42 predictions, with the pooled threshold procedure described in the text.

Model	Run F1	Worst	Collapse	Pooled F1	Brier	ECE
Gated hybrid	0.8908 ± 0.0429	0.8385	0/10	0.8865	0.0257	0.0579
Cross-attention hybrid	0.8894 ± 0.0510	0.8007	0/10	0.8913	0.0267	0.0627
LSTM only (control)	0.8839 ± 0.0436	0.8302	0/10	0.8812	0.0208	0.0548
Weighted-sum hybrid	0.8831 ± 0.0412	0.8062	0/10	0.8906	0.0236	0.0409
Concatenation hybrid	0.8703 ± 0.0543	0.7621	0/10	0.8772	0.0254	0.0560

The fusion ranking reverses. On the fixed partition the gated operator finished last of four and collapsed in one of five seeds; across five trip partitions it finishes *first* of four and

collapses in none of ten runs. Its worst cross-validated run is 0.8385, against 0.0715 on the fixed partition. The concatenation operator, third on the fixed partition, is last here.

Neither ordering should be reported as a result, and we report neither. At the trip level, gating differs from the weighted sum by -0.0040 (95% CI $[-0.0147, +0.0065]$, Holm $p = 0.613$), from cross-attention by -0.0047 ($[-0.0123, +0.0044]$, Holm $p = 0.613$), and from concatenation by $+0.0093$ ($[+0.0019, +0.0199]$, raw $p = 0.023$, Holm $p = 0.070$). **No fusion operator is distinguishable from any other** once the correct unit and multiplicity correction are applied. The four configurations span 0.0205 F1, against a within-configuration standard deviation of 0.04–0.05.

This is the paper's central empirical statement, and it cuts in both directions. It does not vindicate the sample-wise gate: nothing here shows the gate helping. It equally does not condemn it: the fixed-partition result that placed it last, including its collapse, is not reproduced when five partitions are used instead of one. **What the data supports is that the choice of fusion operator in this setting is not resolvable**—and that a study reporting a ranking from a single partition, as the earlier version of this work did, would have reported one of two opposite orderings depending on which partition it happened to use.

The collapse mode of Section 4.3 remains a real and reproducible phenomenon; it recurred at the same seed across a full re-run under a different backend configuration. What these ten cross-validated runs establish is that it is *rare* rather than systematic, and Section 4.7 shows it is conditional on the spatial branch being nearly uninformative. A failure mode that appears in one of fifteen runs is worth documenting for practitioners; it is not evidence that one operator is generally worse than another.

The spatial-branch control is equally instructive: removing the spatial branch entirely yields 0.8839 ± 0.0436 , statistically indistinguishable from every hybrid variant. On the fixed partition the same ablation appeared to cost 3.5 F1 points at $p = 0.035$. We **withdraw that claim**. Section 4.5.1 examines whether the spatial branch helps anywhere at all.

4.5.1. Subtype Stratification

Section 4.1 identified speed-limit violation as the one anomaly subtype whose defining rule depends on a quantity ($v_{\max}$) available only to the spatial branch. Pooling out-of-fold predictions gives 144 such sequences drawn from all 90 trips—enough for a stratified comparison that the single partition, with 13 positives from 14 trips, could not support. Table 14 and Fig. 6 compare LSTM–GAT with static weighted-sum fusion against LSTM-only, with all tests at the trip level and Holm correction across the three subtypes. This comparison assesses the spatial branch within the weighted-sum configuration, not a benefit of sample-wise gating.

Table 14. Subtype stratification on seed-42 pooled cross-validated predictions: static weighted-sum LSTM–GAT versus LSTM-only, using their respective pooled thresholds. Δ denotes their F1 difference (weighted-sum hybrid minus LSTM-only). Intervals and p -values are trip-level over all 90 trips; the Holm column corrects across the three subtypes.

Subtype	n	Weighted-sum hybrid	LSTM-only	Δ	Trip 95% CI	Holm
Speed-limit violation	144	0.1469	0.2312	-0.0843	$[-0.2499, +0.0716]$	0.563
Abnormal dwell	1,154	0.9400	0.9065	$+0.0335$	$[+0.0139, +0.0610]$	0.002
Signal freeze	62	0.0405	0.0773	-0.0367	$[-0.0926, +0.0306]$	0.563

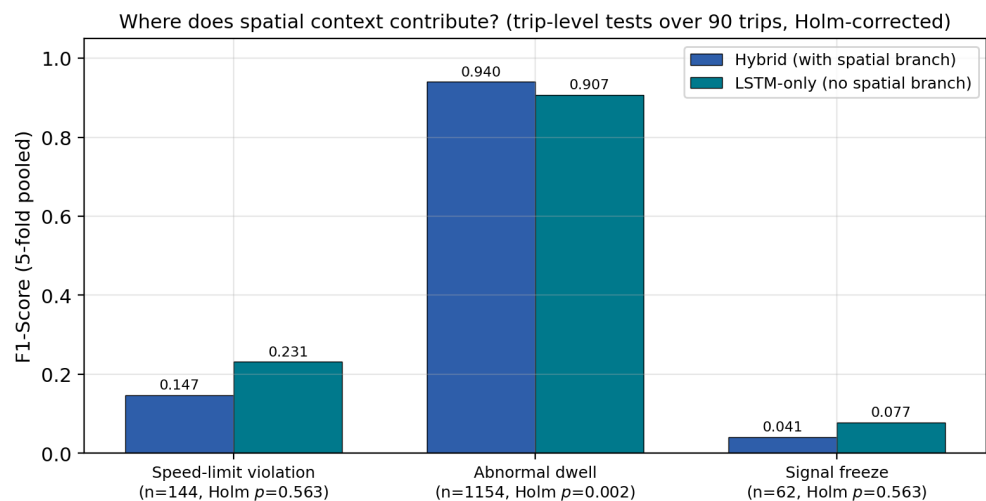


Figure 6. Subtype stratification on pooled cross-validated predictions, comparing static weighted-sum LSTM–GAT with LSTM-only. The spatial branch helps where it was not predicted to (abnormal dwell) and does not help where it holds the decisive feature (speed-limit violation).

This table contains the paper’s one comparison that is significant at the trip level after correction, and it is not the one we set out to test.

The predicted effect is absent. On speed-limit violations—the subtype whose generating rule is a function of $v_{\max}$, a quantity present only in the node features—the hybrid model is *worse* than the temporal-only model (0.1469 versus 0.2312), and the interval spans zero. Both are weak in absolute terms. Although $v_{\max}$ is available to the spatial branch, that branch does not deliver it in a form the classifier uses. Two plausible reasons are that $v_{\max}$ is an imputed default for 49.9% of segments (Section 3.3.1), and that speed-limit violations constitute only 0.5% of rows, providing little gradient signal.

An unpredicted effect is present. On abnormal dwell—by far the most frequent subtype, with 1,154 positives—the static weighted-sum LSTM–GAT model gains +0.0335 F1 over the temporal-only control, with a trip-level interval of [+0.0139, +0.0610] that excludes zero and a Holm-corrected p of 0.002. With 90 clusters rather than 14, the design has enough power to resolve an effect of this size, and it does. We did not predict this subtype and offer an interpretation only tentatively: dwell events are plausibly more discriminable when road context distinguishes a signalized intersection or a congested trunk segment from an open motorway, and that context is exactly what the node features encode. Because it was not hypothesized in advance, we report it as a finding to be confirmed rather than as confirmation.

The two results together locate the spatial branch’s contribution precisely: it is real, it is small, and it is concentrated in the high-frequency subtype rather than in the one whose definition depends on a road attribute. This is a more specific statement than either the earlier version’s claim of a general spatial gain or its subsequent retraction, and it is the statement the data supports.

4.6. Calibration

Table 13 and the reliability diagram in Fig. 7 report probability calibration from the seed-42 pooled out-of-fold probabilities, not an average across the two training seeds. The best-calibrated hybrid is the weighted-sum variant (ECE 0.0409), ahead of concatenation (0.0560), gating (0.0579), and cross-attention (0.0627); the temporal-only control attains the lowest Brier score (0.0208).

We note this as a correction to the earlier version of this work, which reported the gated operator as clearly best calibrated (ECE 0.0338 against 0.0594 for the weighted sum)

on the basis of a single seed per fold. The current archived pooled results reverse that reported ordering. Because the calibration metrics use only seed-42 predictions, this change cannot be attributed to averaging over two seeds, and calibration stability across seeds has not been evaluated here. We therefore do not claim a general calibration advantage for either fusion operator.

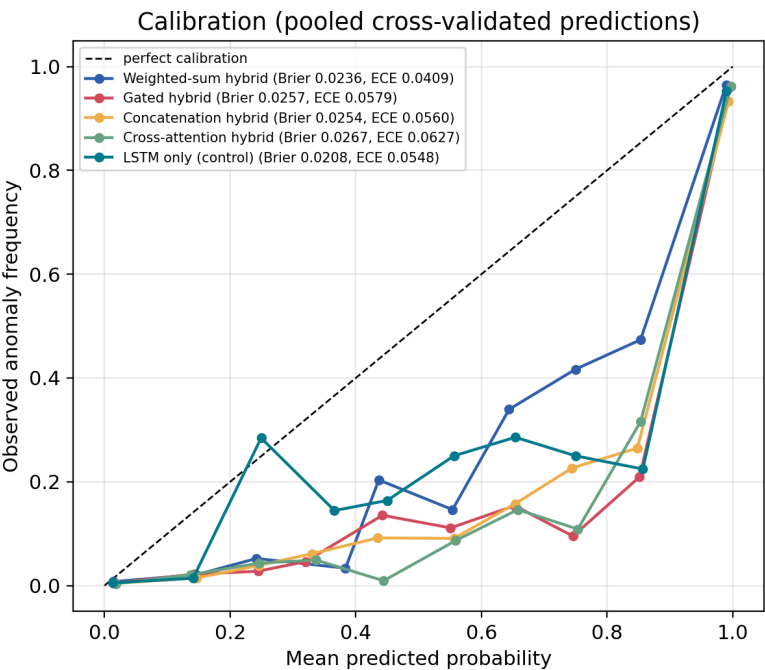


Figure 7. Reliability diagram from pooled cross-validated predictions.

The general point survives the correction even though the specific claim does not. In a monitoring context where alerts are ranked or thresholded by predicted risk, a better-calibrated score is more useful than a marginally higher F1, and calibration is worth reporting for that reason regardless of which variant wins. All five configurations are mildly overconfident, with ECE between 0.041 and 0.063; none is well enough calibrated to be used as a risk score without post-hoc recalibration.

4.7. Graph Construction Ablation

Section 3.3.2 noted that the graph can be built from trip-specific segment instances (4,678 nodes) or merged by physical road (628 nodes). Table 15 reports both.

Table 15. Effect of graph construction, five seeds. The fusion operator is named explicitly in every row: an earlier version of this table labelled its hybrid rows only as “hybrid”, which made them appear inconsistent with the fusion ablation of Table 12 when in fact a different operator had been used.

Configuration	Fusion	Nodes	Deg.	F1-Score	ROC-AUC
GAT only, trip-instance	—	4,678	11.47	0.0767 ± 0.0035	0.5910
GAT only, merged	—	628	1.11	0.0745 ± 0.0047	0.7115
Hybrid, trip-instance	weighted sum	4,678	11.47	0.8713 ± 0.0188	0.9638
Hybrid, merged	weighted sum	628	1.11	0.8413 ± 0.0010	0.9228
Hybrid, trip-instance	gated	4,678	11.47	0.6953 ± 0.3488	0.8752
Hybrid, merged	gated	628	1.11	0.8493 ± 0.0274	0.9684

Merging duplicate nodes substantially improves the *standalone* discriminative power of the spatial branch (ROC-AUC 0.5910 $\rightarrow$ 0.7115), confirming that the duplicate structure degrades message passing. Its F1 nonetheless remains near 0.07, consistent with the label-ambiguity ceiling of Section 3.3.2: several sequences with different labels map to the same node, so no threshold yields good precision and recall simultaneously.

With the fusion operator held fixed, the hybrid is stronger on the trip-instance graph (0.8713 versus 0.8413 for weighted sum), consistent with its higher per-node label purity. The row that repays closer reading is the last one. **The gated operator does not collapse on the merged graph:** it attains 0.8493 ± 0.0274 there, against 0.6953 ± 0.3488 on the trip-instance graph, and records the highest ROC-AUC in the table. This locates the collapse mechanism precisely. The gate fails when it locks onto a spatial branch that carries almost no standalone signal (ROC-AUC 0.5910, barely above chance); on the merged graph the same branch reaches 0.7115, and locking onto it is no longer catastrophic. The instability of Section 4.3 is therefore not a property of gating in the abstract but of gating over an uninformative modality—which is a sharper and more transferable statement than the one we could make from the fixed graph alone.

4.8. Statistical Reassessment at the Trip Level

The window-level p -values retained for the fixed-partition baseline, encoder, and fusion comparisons treat each of the 4,799 test windows as independent. Section 3.4.2 explains why that assumption does not hold: windows overlap by 80%, and windows from the same trip share a driver, a vehicle, and a route. This subsection reassesses those fixed-partition comparisons using the trip as the unit and Holm correction within each family. Its scope does not include the cross-validated fusion comparisons in Section 4.5 or the subtype results in Table 14, which already report trip-level inference over all 90 trips. Table 16 therefore contrasts statistical units for the fixed partition rather than revising every previously reported p -value.

The fixed test partition contains 4,799 windows drawn from **14 trips** (between 75 and 760 windows each, mean 343). Fourteen is the effective sample size for any inference on this partition.

Table 16. Fixed-partition comparisons at the window level and at the trip level. The rule comparator is the one-threshold dwell-only rule, not the three-threshold rule in Table 10. $\Delta F1$ and intervals are computed on the median seed. Trip-level p -values come from sign-flip permutation over trips; the Holm column corrects within each family. Widening is the ratio of trip-level to window-level interval width.

Family	Comparison	p (window)	p (trip)	Holm	Widen.
Baselines	vs. LSTM only	0.549	0.749	1.000	2.43×
	vs. LightGBM	0.302	0.219	0.657	1.81×
	vs. XGBoost	0.049	0.064	0.320	1.80×
	vs. Random Forest	0.754	0.632	1.000	1.07×
	vs. dwell-only rule	< 0.001	0.065	0.320	1.84×
Fusion	gated vs. weighted sum	0.031	0.032	0.097	1.25×
	gated vs. concatenation	1.000	0.500	0.755	1.16×
	gated vs. cross-attention	0.727	0.377	0.755	0.85×
Encoders	vs. adaptive adjacency	< 0.001	0.007	0.056	2.71×
	vs. gated TCN	0.003	0.048	0.335	1.44×
	vs. Chebyshev conv	0.007	0.061	0.365	2.79×
	vs. diffusion conv	0.003	0.062	0.365	2.22×
	vs. GRU	0.119	0.375	1.000	1.71×
	vs. dilated conv	0.115	0.503	1.000	1.19×
	vs. Transformer	0.815	0.439	1.000	1.17×
	vs. no spatial branch	0.549	0.748	1.000	2.43×

The outcome is unambiguous and we report it without softening. **Under the correct statistical unit, none of the sixteen comparisons in Table 16 remains significant after multiplicity correction.** Seven comparisons that were significant at the window level are not significant at the trip level: the advantage over XGBoost, the advantage over the one-threshold dwell-only rule, the gated-versus-weighted-sum difference, and the margins over the STGCN gated temporal convolution, Chebyshev convolution, diffusion convolution, and adaptive adjacency. Confidence intervals widen in most comparisons, but not all; the trip-level to window-level interval-width ratios range from 0.85 to 2.79 across Table 16. The hybrid model's own F1 on this partition, 0.8851 on the median seed, carries a trip-level interval of [0.7231, 0.9470].

Two comparisons illustrate the distinction between descriptive effects and corrected inference. The comparison against adaptive adjacency has a raw trip-level $p = 0.007$ and an unadjusted cluster-bootstrap interval of [+0.679, +0.944] on the median seed, but its Holm-adjusted $p = 0.056$ does not meet the stated 0.05 threshold across the eight encoder comparisons. Neither the large median-seed difference nor the unadjusted interval overrides this corrected result. Across five seeds, adaptive adjacency has low observed mean F1 and high variability (0.4657 ± 0.2833 ; Section 4.2); these are descriptive observations that motivate further stability experiments, not an established collapse mechanism or evidence of statistically significant encoder superiority under the adopted protocol. Similarly, the gated-versus-weighted-sum difference has a raw trip-level $p = 0.032$ and a median-seed F1 difference of -0.038 , but does not survive Holm correction ($p = 0.097$); it does not support a fusion-operator advantage on this partition.

The general lesson is the one this table was constructed to expose. A literature that evaluates overlapping windows as independent observations on a single partition will report significance that a correctly specified test does not support. That is not a criticism of any particular prior study; it applies with full force to the earlier version of this work,

whose central ablation claim rested on a window-level p of 0.035 from one partition. This fixed-partition result is distinct from the cross-validated evidence in Section 4.5, where all 90 trips receive out-of-fold predictions rather than 14, and the fusion and subtype comparisons already use trip-level inference.

4.9. Computational Efficiency and Edge Feasibility

Table 17 and Fig. 8 report parameter counts and single-window inference latency measured with batch size 1, after 50 warm-up iterations, over 500 repetitions on GPU and 60 on CPU. Measurements are taken on the **trip-instance graph** (4,678 nodes), the primary setting of the accuracy tables. The previous version reported latency measured on the merged 628-node graph while reporting accuracy on the trip-instance graph, which understated the cost of the configuration actually being evaluated; the figures below supersede those.

Table 17. Single-window inference latency (median) and parameter counts, measured on the trip-instance graph (4,678 nodes)

Model	Params	GPU	GPU cached	CPU	CPU cached
Hybrid (weighted sum, GAT)	220,802	2.87 ms	0.64 ms	27.82 ms	7.03 ms
Hybrid (gated, GAT)	224,962	2.80 ms	0.78 ms	25.07 ms	7.63 ms
Hybrid (concat)	222,849	2.56 ms	0.55 ms	23.14 ms	7.32 ms
Hybrid (cross-attention)	237,441	2.91 ms	0.96 ms	19.83 ms	6.61 ms
LSTM only	215,234	0.80 ms	0.64 ms	7.69 ms	7.03 ms
GRU + GAT	170,626	2.60 ms	0.66 ms	33.63 ms	8.58 ms
Gated TCN + GAT	122,242	2.76 ms	0.88 ms	24.79 ms	1.83 ms
Dilated conv + GAT	416,258	4.22 ms	2.26 ms	27.42 ms	5.98 ms
Transformer + GAT	292,098	3.30 ms	1.30 ms	22.48 ms	2.84 ms
Adaptive adjacency + LSTM	321,850	6.18 ms	0.69 ms	55.40 ms	6.85 ms

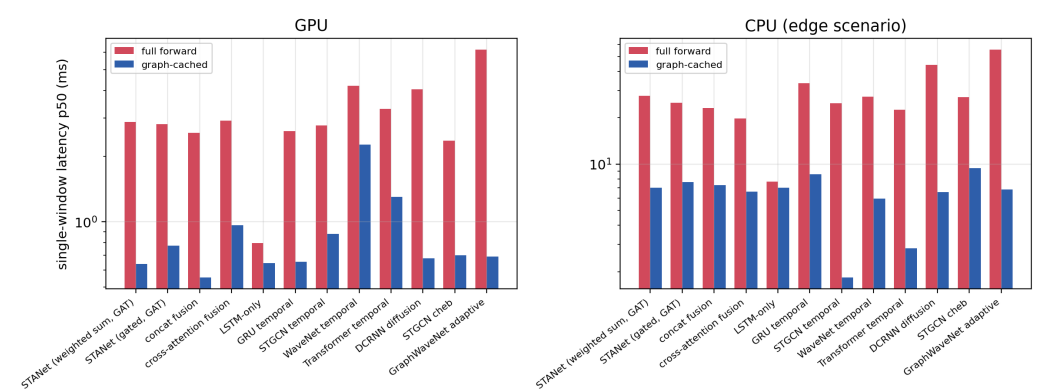


Figure 8. Single-window latency with and without graph-embedding caching, on GPU and CPU (log scale).

The primary model contains 220,802 parameters (≈ 0.9 MB in fp32), of which the temporal branch accounts for 200,704 (90.9%), the projection layers 12,416, the classifier 2,113, the spatial branch only 5,568 (2.5%), and the fusion operator a single scalar. Peak GPU memory is 66 MB.

Despite that small share, the spatial branch dominates inference cost, because a naive implementation recomputes embeddings for all 4,678 nodes on every forward pass. **Since the road network is static, node embeddings can be computed once and cached**, reducing GPU latency from 2.87 to 0.64 ms ($4.5\times$) and CPU latency from 27.82 to 7.03 ms ($4.0\times$).

For static weighted-sum LSTM–GAT, the median cached latency equals the corresponding LSTM-only value at the reported precision on the measured GPU and CPU (0.64 ms and 7.03 ms, respectively). This agreement is specific to that configuration, the tested hardware, and the stated measurement protocol; it does not establish zero inference overhead in general. The other fusion operators have different cached latencies in Table 17, so the equality does not extend to them.

At 27.82 ms per uncached window on CPU ($p_{95} = 39.92$ ms), or 7.03 ms cached ($p_{95} = 8.77$ ms), a single-core edge device could process on the order of 35 windows per second without caching and 140 with it, against a one-second sampling resolution. Caching is therefore not an optimization but a precondition for single-core deployment at scale. Memory is not a constraint at this model size. These figures characterize computational feasibility only; they do not address detection quality under real sensing conditions.

4.10. Scope of the Contribution

It is worth stating directly what kind of claim this paper does and does not make, because the two are easily conflated in this literature.

This is an *evaluation study*. Its object is a set of design decisions—encoder pairing, fusion operator, graph construction—and its result is a statement about how much evidence a controlled comparison can actually support for each of them. It is not a report of a deployable traffic anomaly detector, and none of the numbers here should be read as an estimate of field performance. The data are simulated; the labels are, for most subtypes, threshold functions of the model’s own inputs; the road network is one city; the trip count is 90. Each of these independently rules out a deployment claim, and we make none.

What survives those limitations is the methodological finding, and it does so precisely because it is a *relative* statement evaluated under a single fixed protocol: a sample-wise gate that is presented as the architectural contribution of a hybrid model does not, on this data, outperform a single learnable scalar, and the difference between the two is smaller than the variation across random seeds. That comparison does not depend on the data being real, because every variant faces the identical task, partition, budget, and decision rule. It does depend on the comparison being controlled, which is why Section 3.4.1 and Section 3.4.2 are given the space they are.

Whether the same conclusion holds on field-collected trajectories is an open question that this dataset cannot answer, and we identify it in Section 5.3 as the decisive next experiment rather than implying it here.

4.11. Threats to Validity

Synthetic data. All findings are obtained on simulated trajectories over one city network. No real sensor data, verified incident record, or prospective stream is involved, and no claim about real-world detection performance is made.

Label–input dependence. Most anomaly subtypes are threshold functions of the model’s own inputs (Section 4.1). Conclusions about *relative* architectural merit are unaffected, since all configurations face the same task, but absolute performance figures reflect a partly self-referential problem and should not be compared to results on independently labelled data.

Statistical power. This is the tightest constraint on what the paper can claim, and it is a consequence of the sampling design rather than of the analysis. Because windows overlap by 80% and share a trip, the unit of analysis is the trip (Section 3.4.2), and the dataset contains only 90 trips: 14 of them in the fixed test partition, all 90 across the cross-validation folds. Interval estimates on the fixed partition are therefore wide, and comparisons that appear significant at the window level frequently are not once the correct unit is used. We

report both, labelled, and base conclusions on the trip-level, Holm-corrected results for the corresponding evaluation regime, with cross-validation providing the primary evidence where available. Differences that the design cannot resolve are presented as point estimates with intervals and are not claimed as significant—including differences that would favour the architecture proposed in the earlier version of this work.

Encoder adaptation. STGCN, DCRNN, and Graph WaveNet are represented by their core operators rather than their full published architectures. This is necessary given the task mismatch, but the results should be read as evidence about those operators in this setting rather than as a benchmark of the published systems.

Imputation dependence. Roughly half of the lane counts and speed limits are author-defined defaults, which directly affects the node features available to the spatial branch. A sensitivity analysis over these defaults was not performed.

5. Conclusion and Future Work

This study evaluated the components of a hybrid LSTM–GAT traffic anomaly detector under a controlled protocol in which one element is varied at a time and every result is reported over multiple seeds with interval estimates and paired tests.

5.1. Summary of Findings

- **The fixed partition supports no claim at all.** It holds 4,799 windows but 14 trips. At the trip level, after Holm correction, none of sixteen comparisons is significant—including seven that are significant when windows are treated as independent. The hybrid model's own F1 on that partition carries an interval of $[0.7231, 0.9470]$.
- **The fusion ranking is partition-dependent, in both directions.** The sample-wise gate ranks last of four on the fixed partition and first of four under cross-validation, where no operator differs significantly from any other. We therefore make no accuracy claim for or against the gating mechanism, and we withdraw the earlier claim against it.
- **A reproducible collapse mode exists and is conditional.** The gate collapsed to $F1 = 0.0715$ at one of five seeds on the fixed partition, recurring at the same seed in an independent re-run under a different backend configuration. It occurred in none of ten cross-validated runs, and it disappears on a graph construction where the spatial branch carries more standalone signal. It is a rare, conditional failure mode worth documenting for practitioners—not evidence of general inferiority.
- **The spatial branch helps, but not where predicted.** Static weighted-sum LSTM–GAT improves abnormal-dwell detection over LSTM-only by $+0.0335$ F1 (trip-level 95% CI $[+0.0139, +0.0610]$, Holm $p = 0.002$); speed-limit violation, the subtype whose rule depends on $v_{\max}$, shows no benefit. This is the study's one properly powered positive result, and it was not hypothesized in advance.
- **Encoder rankings remain descriptive.** LSTM–GAT has the highest observed mean F1 in the fixed-partition encoder benchmark, and the LSTM, GRU, and Transformer configurations lie within 0.0015 F1 of one another. None of the eight encoder comparisons remains significant after trip-level Holm correction (Table 16). Thus, these results establish neither spatial-operator nor temporal-encoder superiority; they also do not demonstrate equivalence.
- **Gate interpretability is seed-dependent.** The class-conditional shift in g_{gate} varies in magnitude and sign across seeds and does not constitute a stable model property.
- **Caching reduces measured inference cost** (Section 4.9). For static weighted-sum LSTM–GAT, caching static node embeddings reduces single-window latency by $4.5\times$ on GPU and $4.0\times$ on CPU. Its median cached latency matches LSTM-only at the

reported precision on the tested hardware; this equality is not established for other fusion operators or hardware.

- **Simple references are strong.** A zero-parameter three-threshold rule reaches $F1 = 0.8536$, while Random Forest on window aggregates has a mean $F1$ only 0.0014 below that of static weighted-sum LSTM–GAT, with higher ROC-AUC.

5.2. Implications

The limitations bounding these conclusions are collected in Section 4.11. The broader implication is methodological, and it has three parts that we would separate.

The unit of analysis is not a detail. Sliding-window evaluation is near-universal in this literature, and windows cut with a stride shorter than their length are not independent observations. Reporting window-level p -values under that design does not slightly overstate significance; in our case it turned sixteen non-significant comparisons into seven significant ones. Any study using overlapping windows should state its cluster structure and test at the level of the independent unit—here the trip, elsewhere the vehicle, the sensor, or the day.

One partition cannot order architectures. We report the same four fusion operators ranked in opposite orders on one partition and across five, with the mechanism under study finishing both last and first. Neither ordering is wrong; both are under-determined. A single grouped partition, even a well-constructed one, is a sample of size one from the space of partitions, and architectural rankings are not stable with respect to that sample.

Error bars must be earned. Extending the encoder benchmark from three seeds to five raised the LSTM–GAT standard deviation from 0.0033 to 0.0188 and changed which comparisons met the window-level significance threshold. None of the eight encoder comparisons remains significant after trip-level Holm correction. A tight standard deviation over three runs is not sufficient evidence of stability, and increasing the number of seeds does not make dependent windows independent statistical units.

We would additionally encourage reporting a rule-based or classical reference whenever labels are generated by known rules, since it helps contextualize the performance of learned models. On the fixed partition, the zero-parameter three-threshold rule reaches $F1 = 0.8536$, compared with mean $F1 = 0.8713$ for static weighted-sum LSTM–GAT: a difference of $0.8713 - 0.8536 = 0.0177$ using the reported rounded values. This comparison is specific to the weighted-sum configuration and does not extend to every neural configuration tested.

5.3. Future Directions

- **Dataset redesign.** Sharing road segments across trips and increasing the prevalence of context-dependent subtypes such as speed-limit violation would allow the spatial branch's contribution to be tested with adequate power—the decisive experiment that the present data cannot support.
- **Simulation-to-real transfer.** Future work should evaluate whether domain adaptation can transfer representations learned from physics-informed synthetic trajectories to field-collected data. The temporal and spatial branches should be assessed separately to determine whether adaptation improves cross-city performance.
- **Stabilized gating.** Since the collapse mode arises from the multiplicative form of Eq. 16, entropy regularization on g_{gate} , a bounded gating range, or an auxiliary supervision signal on the gate are candidate remedies worth evaluating.
- **Imputation sensitivity.** Road-type-specific lane and speed-limit defaults should be varied systematically to quantify how strongly the spatial branch depends on author-defined assumptions.

- **Real-world and cross-city validation.** Field-collected sensor data, independently verified incidents, prospective streams, and external road networks are required before any claim of deployment suitability. Processing such high-frequency streams will also require efficient trajectory preprocessing, e.g., via classical simplification methods (Ramer–Douglas–Peucker [32]; Meratnia and de By’s time-ratio extension [33]; Potamias et al.’s spatiotemporal stream sampling [34]) or more recent artificial neural network (ANN)-based [35] and window-based [36] reduction approaches, to limit computational overhead. Their effects on noise-driven false positives and the preservation of short-duration anomalies should be evaluated before deployment.

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Informed Consent Statement: Not applicable. No human participants or real-world vehicle trajectory data were involved in this study.

Data Availability Statement: The synthetic dataset, its data card, the complete preprocessing and modelling code, the experiment scripts, every result file reported in this paper, and the script that verifies the manuscript’s numbers against those files are available at <https://github.com/keremkartal/Anomaly-Detection> and permanently archived at doi:10.5281/zenodo.22637361, which resolves to the exact snapshot (v2.0.0) that produced every number reported here. The concept identifier doi:10.5281/zenodo.22637360 resolves to the most recent version. No real-world vehicle trajectory data was used.

The repository contains two mechanisms that support the internal consistency of the tables reported here. First, every result file carries a *protocol stamp*: a hash over the seed set, epoch budget, patience, batch size, learning rate, scheduler, dropout, checkpoint criterion, threshold rule, cuDNN flags, window length, stride, split seed, and library versions. Scripts that consume another script’s results assert that the stamps match and abort otherwise. Second, configurations that appear in more than one table are trained once and cached under their protocol stamp, so those tables read the same run rather than two independently trained runs that are assumed to agree. Section 3.4.1 describes why these mechanisms were introduced.

Because several anomaly subtypes are defined as threshold functions of the model’s input channels, a zero-parameter rule-based classifier is reported alongside all learned models as a reference point.

Conflicts of Interest: The authors declare no conflicts of interest.

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